YASIN BOLOORCHI

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EDUCATION

Ph.D. in Computer Science, University of Iowa — 2022 – May 202
Focus: Distributed systems, wireless networks, and system reliability.

B.S. in Computer Engineering, University of Kurdistan — 2016 – 2021
Bachelor's Thesis: Design and implementation of an IoT-based demand-side management system using big-data technologies.

TECHNICAL SKILLS

Languages	Python, C, Bash
Systems	Linux, Distributed Systems, Networking, Docker, Kubernetes
Infrastructure	FlexRIC, srsRAN, Open5GS, O-RAN/xApps
Tools & Methods	Git, TLA+, Model Checking, Protocol Analysis
Other	LaTeX, Web Scraping

EXPERIENCE

Research Engineer / Software Developer — 2022 – Present
University of Iowa — *Iowa City, IA*

- Designed and implemented simulation and analysis tools to evaluate predictability and schedulability of wireless network protocols.
- Built symbolic-execution-based tooling to analyze timing behavior and correctness of low-level network code.
- Developed and maintained reproducible experimentation pipelines for an NIST-based O-RAN testbed using Python, C, and Linux-based environments.

Software Engineering Intern — Summer 2020
Kurdistan Cloud Computing Development Company — *Kurdistan, IRN*

- Developed a traffic management and reporting system for organizational use.
- Integrated Telegram Messenger APIs to automate employee coordination and reporting workflows.

PROJECTS

Wireless Protocol Code Simulator. Built a symbolic-execution-based simulator to analyze timing behavior and schedulability of wireless protocol implementations, emphasizing correctness, repeatability, and performance evaluation.

IoT Energy Consumption Analytics Platform. Designed a data pipeline for ingesting and analyzing IoT-generated power consumption data using statistical optimization techniques and big-data tooling.

Full-Text Search Engine for Persian Poetry. Implemented a scalable search engine supporting efficient query-based retrieval across large text corpora.

PUBLICATIONS

Y. Boloorchi, S. Azizi. "An IoT–Fog–Cloud Framework for Demand-side Management in Smart Grids." International Conference on Internet of Things and Its Applications (IoT 2021).